

# MD SAIF ALAM

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## PROFESSIONAL SUMMARY

Final-year B.E. AI and Machine Learning student and Samsung PRISM intern with hands-on experience building production-grade, AI-powered full-stack applications. Proficient in Python, FastAPI, React, RAG pipelines, LLMs, Generative AI, and Computer Vision with proven results across fraud detection, malware analysis, counterfeit IC detection, and GenAI systems. Experienced team lead seeking AI Engineering or Full Stack Development roles to deliver scalable, intelligent, production-ready solutions.

## TECHNICAL SKILLS

**Languages:** Python, C, JavaScript, HTML5, CSS3, SQL

**Frameworks:** React, FastAPI, Flask, PyTorch, Scikit-Learn, LangChain, Pandas, NumPy, Pydantic, Tailwind CSS

**AI / GenAI:** RAG Pipelines, LLMs, Generative AI, Prompt Engineering, Hugging Face, YOLO, Ollama, Vector Embeddings, MediaPipe, AI Agents

**Databases:** PostgreSQL, MongoDB, MySQL, QdrantDB

**Tools and Cloud:** Git, Docker, REST APIs, AWS S3, Vercel, Render, Uvicorn, Jupyter Notebook

## EXPERIENCE

**Project Intern** | Samsung R&D Institute India — Samsung PRISM

Jan 2026 – July 2026

- Leading a 4-member team building a production-grade **AI-based UE Capability Logs Extractor and Summarizer** using RAG pipelines and LLM integration to parse NR, E-UTRA, and CR band data.
- Developing scalable full-stack solution with **FastAPI backend and React + Vite frontend** in collaboration with Samsung engineers on real-world 5G/LTE data extraction problems.

## PROJECTS

**ThreatLens — APK Malware Analysis Platform** — *FastAPI, React, Groq LLM, Androguard, Python* [GitHub](#) | [Live Demo](#)

- Built full-stack APK malware analysis tool with Androguard metadata extraction, heuristic risk scoring, and malware fingerprint matching; both frontend and backend deployed on Render.
- Architected a **4-agent Generative AI pipeline** delivering plain-English verdicts in under 30 seconds with automated PDF report generation.

**MuleNet — AI Fraud Intelligence Platform** — *FastAPI, React, LightGBM, XGBoost, SHAP, Scikit-Learn* [GitHub](#) | [Live Demo](#)

- Engineered ensemble Machine Learning pipeline achieving **AUC-ROC 99.1%, Precision 98.7%, 96.3% High/Critical detection** via 5-fold cross-validation; integrated SHAP explainability.
- Deployed a **sub-50ms live scoring REST API** for real-time fraud detection at scale.

**SiliconSeal — IC Counterfeit Detection System** — *FastAPI, React, YOLOv8, PaddleOCR, Groq LLM, OpenCV* [GitHub](#) | [Live Demo](#)

- Built for **BEL at SIH 2025 Finals** — YOLOv8 IC detection → PaddleOCR extraction → Aho-Corasick pattern matching → weighted scoring engine (Part 60%, Date 25%, Lot 15%).
- Integrated **Groq LLM (Llama 3.3 70B)** for plain-English risk assessment supporting single, batch, and live camera scanning modes across 27+ IC entries in knowledge base.

**ConnectHub — Scalable Full Stack Social Platform** — *FastAPI, React 19, PostgreSQL, SQLAlchemy, AWS S3, Docker* [GitHub](#)

- Architected fully **async RESTful API** with JWT authentication, Argon2 hashing, email-based password reset, **AWS S3** storage, Alembic migrations, and Docker containerization.

## EDUCATION

**Cambridge Institute of Technology, Bangalore**

Expected May 2027

*B.E. in Artificial Intelligence and Machine Learning* | CGPA: **8.6 / 10**

**Class XII (CBSE) | Hem Sheela Model School, Durgapur, WB**

2022 | 89.2%

**Class X (CBSE) | Eastern Railway High School, Asansol, WB**

2020 | 95.2%

## CERTIFICATIONS

- Oracle — [OCI 2025 Generative AI Professional](#) — 2025
- Oracle — [Oracle AI Vector Search Professional](#) — 2025
- Oracle — [OCI 2025 AI Foundations Associate](#) — 2025

## ACHIEVEMENTS AND LEADERSHIP

- 2nd Place, MLH Hackathon** — Major League Hacking Hackathon at Atria Institute of Technology, Bangalore.
- Smart India Hackathon (SIH) 2025 Finalist** — National level finalist representing BEL for IC counterfeit detection solution.
- Team Lead — Samsung PRISM and National Hackathons** — Led 4-member cross-functional teams across multiple national-level competitions and Samsung R&D internship program.